

Python Programming

Session 1

Input() and Print():

- 1) Declare a variable X, assign to it an initial value, increment it by 1 and show the result.
- 2) Use the function "input" to assign a value to a variable.
- 3) Use both, "input" and "print" to display the sentence "Select a number" on the screen.
Then display the selected number with the following format:
"The number is: X"
- 4) Print two sentences separated by a blank line. Can you do it with only one call to "print"?

'int()' i 'float()':

- 5) Ask the user to introduce 2 numbers (height and width) and show the area of the corresponding rectangle. Which data type do you expect?
- 6) Ask the user to introduce the radius of a circle and show its area on the screen.

'if', 'if..else' i 'elif':

- 7) Create a menu where one can choose between a circle or a rectangle. Display the selected option. Something like this:

```
1-Rectangle
2-Circle
Select option:
```

9.1) Make it display "Incorrect option" if the user selected a number different from one or two.

- 8) Depending on the selected option, ask for the necessary inputs and display the area.
- 9) Create a BMI calculator
$$\text{BMI} = \text{weight}(\text{kg}) / \text{height}(\text{m}^2)$$
 - a. If $\text{BMI} < 18.5$ show "Low weight"
 - b. If $\text{BMI} > 18.5$ i $\text{BMI} \leq 25$ show "Normal Weight"
 - c. If $\text{BMI} > 25$ i $\text{BMI} \leq 30$ show "Overweight"
 - d. If it is bigger than 30 show "Obesity"
- 10) Introduce two numbers and compare if they are equal. Display the result.
- 11) Introduce a person's age and verify if it's an adult or not.